

Subarachnoid Hemorrhage with Increased ICP



Section I: Scenario Demographics

Scenario Title:	Subarachnoid Hemorrhage with Increased Intracranial Pressure
Date of Development:	21/12/2014 (DD/MM/YYYY)
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To explore treatment options for the patient presenting to the emergency department with a neurologic catastrophe.
CRM Objectives:	1) Anticipate and appropriately plan for intubation in the patient with possible increased intracranial pressure. 2) Ensure a shared mental model by summarizing the case to team members at key moments.
Medical Objectives:	1) Recognize the importance of and appropriately manage uncontrolled hypertension in the setting of a suspected non-traumatic SAH. 2) Consider the effect of intubation on intracranial pressure in the setting of a CNS catastrophe and options to mitigate this effect. 3) Apply therapeutic strategies to manage increased intracranial pressure in the setting of acute hydrocephalus.

Case Summary: Brief Summary of Case Progression and Major Events

A 45-year-old male who suffered an aneurysmal subarachnoid hemorrhage while weightlifting presents to the emergency department requiring intubation for airway protection and develops acute hydrocephalus requiring ICP lowering maneuvers before definitive surgical management.

References

- 1) Marx, J., Hockberger, R., & R. M., W. (2013). Headache Disorders. In *Rosen's Emergency Medicine: Concepts and clinical practice* (8th ed., pp. 1390-1391). Philadelphia, Pa.: Elsevier Saunders.
- 2) Diring, M., Bleck, T., Menon, D. et al. (2011) Critical Care Management of Patients Following Aneurysmal Subarachnoid Hemorrhage: Recommendations from the Neurocritical Care Society's Multidisciplinary Consensus Conference. *Neurocrit Care.* Sep; 15(2):211-40
- 3) Connolly, E., Rabinstein, A., Derdey, C. et al. (2012) Guidelines for the management of aneurysmal subarachnoid hemorrhage: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke.* 2012 Jun;43(6):1711-37



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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You are working an evening shift at a tertiary care centre emergency department with full surgical capabilities. A patient is brought into the resuscitation area by ambulance with decreased mental status.			
B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin ☐ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☐ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates	Brief Description of Role		
N/A	N/A.		
C. Required Monitors			
☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☒ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
E. Moulage			
Gym attire for mannequin.			
F. Approximate Timing			
Set-Up:	5 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Patient Profile and History			
Patient Name: Philip Yates		Age: 45	Weight: 100kg
Gender: ☒ M ☐ F		Code Status: Unknown	
Chief Complaint: Altered Mental Status			
History of Presenting Illness: The patient was at the gym lifting weights; he complained of a headache to his friend and suddenly fell to the ground. The patient remained unconscious with sonorous breathing. The ambulance was called.			
Past Medical History:	HTN Dyslipidemia T2DM	Medications:	Bisoprolol 5mg PO QD Atorvastatin 40mg PO QD Metformin 500mg PO BID
Allergies: NKDA			
Social History: ½ PPD smoker, recreational EtOH, no Hx of illicit drug use.			
Family History: Non-contributory			
Review of Systems:	<Unable to obtain review of systems history from patient>		
B. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☒ Monitor ON, no data displayed	☐ Monitor ON Standard Display
HR: 110/min	BP: 198/102	RR: 8/min	O ₂ SAT: 95 %
T: 36.9°C	Glucose: 10 mmol/L	GCS: 7 (E2-pain V2-sounds M4-withdraws)	
General Status: Altered, somnolent, decreased responsiveness			
CNS:	Pupils 3mm bilaterally, minimally responsive. Gaze preference to right. Unable to test upper & lower extremity strength/sensation, but not moving left upper extremity. +Babinski reflex on left. 3+ reflexes on left side.		
HEENT:	Normal.		
CVS:	Tachycardia, no extra heart sounds, normal peripheral pulses.		
RESP:	Bradypneic, GAEB		
ABDO:	Normal		
GU:	Normal		
MSK:	Normal, neurologic findings as above	SKIN:	Normal

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus Tach HR: 110/min BP: 200/102 RR: 8/min O ₂ SAT: 95 % T: 36.9°C	GCS: 7 (E2-pain V2-sounds M4-withdraws) Minimally responsive/somnolent	Learner Actions - ☐ Monitors/Full vitals - ☐ Physical exam (+Neuro) - ☐ IV Access - ☐ Laboratory Investigations - ☐ IV anti-hypertensive - ☐ EKG - ☐ Pre-treatment - ☐ Induction medication - ☐ Paralytic medication - ☐ Intubation	Modifiers <i>Changes to patient condition based on learner action</i> - IV anti-hypertensive 1 st bolus → Δ BP to 187/106 - IV anti-hypertensive 2 nd bolus or infusion → Δ BP to 155/95 Triggers <i>For progression to next state</i> - Intubation → 2.Intubation - 5 minutes → 3.Hydrocephalus
2. Intubation O ₂ SAT → 99% RR → 12 Display (if possible) - ETCO ₂ = 48 - FiO ₂ ; 1.0 - T _v : 500mL - PEEP: 5	Pt paralyzed post intubation and sedated.	Learner Actions - ☐ Confirmation of tube - ☐ OG tube - ☐ Post-intubation CXR	Modifiers Triggers - 2 minutes → 3.Hydrocephalus
3. Hydrocephalus HR → 55 over 1 minute BP → 220/112	Posturing movements (if succinylcholine used as paralytic agent) Right pupillary dilatation	Learner Actions - ☐ Elevate head of bed - ☐ Hyperventilation - ☐ Mannitol 1gm/kg of 20% = 500ml - ☐ Hypertonic Saline 3% 250ml bolus - ☐ Neurosurgical consult for EVD	Modifiers - Hyperventilation → ΔETCO ₂ to 35 and ΔRR accordingly Triggers - ≥2 ICP lowering methods → 4.Resolution
4. Resolution HR → 80 over 30 seconds BP → 160/96		Learner Actions - ☐ CT head - ☐ ICU consultation - ☐ ± Nimodipine 60 mg OG - ☐ ± Seizure prophylaxis - ☐ ± Tranexamic Acid 1g IV	END SCENARIO

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results					
VBG	pH: 7.29	PCO ₂ : 62	PO ₂ : 45	HCO ₃ : 23	Lactate: 1.8

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☐ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What made the intubation especially important in this case? 2) What are examples of optimal times to summarize the case for your team? 3) What are risk factors for rebleeding in non-traumatic SAH? 4) What is the target systolic BP in SAH? What can you use to achieve this target? 5) What are signs of acute hydrocephalus? 6) List management strategies for acute hydrocephalus	
Key Moments	
1) Peri-intubation	
2) Cushing's response for increased ICP	
3) Prioritizing options in ICP management; what to give first?	